

JAWAHARLAL NEHRU NEW COLLEGE OF ENGINEERING

DEPARTMENT OF ROBOTICS AND ARTIFICIAL INTELLIGENCE

MACHINE LEARNING ASSIGNMENT

Iranian Telecom Customer Churn Dataset

Submitted By:

NAME: SWAROOP M G

USN: 4JN24RI054

Guided By:

DEEPANKAR V K

Assistant Professor Department of Robotics and Artificial
Intelligence

JNNCE, Shivamogga

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ABSTRACT

Customer churn prediction is an important task in the telecommunications industry, as retaining existing customers is often more cost-effective than acquiring new ones. The objective of this project is to develop machine learning models capable of predicting whether a customer is likely to discontinue a telecom service based on historical customer behavior and service usage patterns.

The Iranian Telecom Churn Dataset was used for this study, containing customer information such as call failures, complaints, subscription length, charge amount, frequency of use, SMS frequency, customer value, and other service-related attributes. Data preprocessing techniques including data cleaning, train-test splitting, and feature scaling were applied to prepare the dataset for model development.

Three classification algorithms, namely Logistic Regression, Decision Tree, and Random Forest, were implemented and evaluated. Model performance was assessed using accuracy, precision, recall, F1-score, confusion matrix analysis, and cross-validation techniques. Hyperparameter tuning was performed to improve model performance and reduce overfitting.

The comparative analysis identified the most effective model for customer churn prediction. The findings demonstrate how machine learning can assist telecommunication companies in proactively identifying customers at risk of churn and implementing targeted retention strategies to improve customer satisfaction and business profitability.

1. INTRODUCTION

In today's highly competitive telecommunications industry, customer retention has become a significant challenge for service providers. Customers frequently switch to competing providers due to factors such as service quality issues, pricing concerns, lack of customer support, and better offers from competitors. This phenomenon, commonly known as customer churn, can result in substantial revenue loss and increased customer acquisition costs for organizations.

Customer churn prediction involves identifying customers who are likely to discontinue a service in the near future. By accurately predicting churn, companies can take preventive measures such as personalized offers, loyalty programs, and improved customer support to retain valuable customers.

Machine learning techniques have emerged as powerful tools for analyzing customer behavior and predicting future outcomes. By leveraging historical customer data, predictive models can uncover patterns and relationships that indicate churn risk. These insights enable businesses to make data-driven decisions and improve customer retention strategies.

This project focuses on developing a customer churn prediction system using the Iranian Telecom Churn Dataset. The dataset contains customer-related information such as service usage, subscription details, complaints, customer value, and communication patterns. Multiple machine learning algorithms, including Logistic Regression, Decision Tree, and Random Forest, were implemented and evaluated to determine the most suitable model for churn prediction.

The primary objectives of this project are:

- To analyze customer behavior and identify factors influencing churn.
- To preprocess and prepare the dataset for machine learning applications.
- To develop and evaluate multiple classification models.
- To compare model performance using various evaluation metrics.
- To identify the most effective model for customer churn prediction.
- To provide business recommendations that can help telecommunication companies improve customer retention.

The outcome of this project demonstrates the practical application of machine learning techniques in solving real-world business problems and highlights the importance of predictive analytics in customer relationship management.

2. DATASET DESCRIPTION

2.1 Dataset Overview

The dataset used in this project is the Iranian Telecom Churn Dataset. This dataset was collected from an Iranian telecommunications company over a period of twelve months and is widely used for customer churn prediction research.

The dataset contains customer behavioral and service usage information that can be utilized to predict whether a customer is likely to discontinue the telecom service. The target variable, Churn, represents the customer's status at the end of the observation period.

Dataset Characteristics

Attribute	Value
Dataset Name	Iranian Telecom Churn Dataset
Domain	Telecommunications

Attribute	Value
Total Records	3150
Total Features	13
Target Variable	Churn
Problem Type	Binary Classification
Data Type	Multivariate
License	CC BY 4.0

2.2 Dataset Features

The dataset contains the following attributes:

Feature Name	Description
Call Failure	Number of failed calls made by the customer
Complains	Indicates whether the customer submitted complaints
Subscription Length	Duration of customer subscription
Charge Amount	Total service charges incurred
Seconds of Use	Total duration of service usage
Frequency of Use	Frequency of service utilization
Frequency of SMS	Number of SMS messages used
Distinct Called Numbers	Number of unique contacted phone numbers
Age Group	Customer age category
Tariff Plan	Type of tariff plan subscribed
Status	Customer service status
Age	Customer age
Customer Value	Estimated value of the customer to the company
Churn	Target variable indicating whether the customer churned or not

2.3 Target Variable

The target variable in this project is Churn.

- Churn = 0 : Customer remains with the company.
- Churn = 1 : Customer leaves the company.

The objective of the machine learning models is to accurately classify customers into these two categories based on historical customer behavior and service usage information.

2.4 Importance of the Dataset

Customer churn prediction is a critical business application because retaining existing customers is generally less expensive than acquiring new customers. By analyzing customer behavior patterns and identifying churn risk factors, telecommunication companies can take proactive actions to improve customer retention and customer satisfaction.

The Iranian Telecom Churn Dataset provides valuable insights into customer interactions, usage behavior, and service engagement, making it suitable for developing predictive machine learning models.

3. EXPLORATORY DATA ANALYSIS (EDA)

3.1 Introduction

Exploratory Data Analysis (EDA) is an important step in the machine learning workflow that helps in understanding the characteristics of the dataset before model development. EDA enables the identification of patterns, trends, relationships, and potential anomalies within the data.

The primary objective of EDA in this project was to understand customer behavior, analyze feature distributions, examine relationships among variables, and identify factors that may influence customer churn.

3.2 Dataset Overview

The dataset contains customer information related to telecom service usage, complaints, subscription details, customer value, and communication patterns. Before model development, the dataset was visually and statistically analyzed to gain meaningful insights.

3.3 Churn Distribution Analysis

The distribution of the target variable (Churn) was analyzed to understand the balance between churned and non-churned customers.

A count plot was generated to visualize the number of customers who remained with the company and those who discontinued the service.

Observation

The majority of customers belonged to the non-churn category, while a smaller proportion represented churned customers. This indicates the presence of class imbalance, which is common in customer churn datasets.

3.4 Feature Distribution Analysis

Histograms and distribution plots were used to examine the distribution of individual features.

The analysis helped identify:

- Customer usage patterns
- Complaint frequency trends
- Distribution of customer value
- Subscription length variations
- Service usage behavior

Observation

Several features exhibited varying distributions, indicating differences in customer engagement and service utilization patterns.

3.5 Correlation Analysis

A correlation matrix was generated to examine relationships among numerical variables.

The correlation heatmap provided a visual representation of the strength and direction of relationships between features.

Observation

The analysis revealed that certain variables exhibited moderate to strong correlations, indicating potential relationships among customer behavior attributes. Features such as service usage, frequency of use, customer value, and charge amount showed meaningful associations.

The target variable also demonstrated varying degrees of correlation with several customer-related attributes, suggesting their potential importance in churn prediction.

3.6 Customer Behavior Insights

EDA revealed several important business insights:

- Customers with frequent complaints appeared more likely to churn.
- Lower service usage was associated with increased churn probability.
- Customer value showed a relationship with customer retention.
- Subscription duration influenced customer loyalty and engagement.
- Service interaction metrics played a significant role in customer behavior patterns.

3.7 Importance of EDA

The insights obtained from EDA provided a deeper understanding of customer behavior and helped identify relevant features for predictive modeling. These findings guided the subsequent preprocessing and machine learning stages of the project.

3.8 Summary

Exploratory Data Analysis successfully uncovered patterns and relationships within the dataset. The analysis highlighted key factors associated with customer churn and provided valuable business insights. These findings established a strong foundation for data preprocessing and machine learning model development.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Ignore warnings
import warnings
warnings.filterwarnings('ignore')
```

EXPLANATION:

Software Libraries Used

The project was implemented using Python and several data science libraries that support data manipulation, visualization, and machine learning tasks.

Pandas

Pandas is a powerful data analysis library used for loading, cleaning, manipulating, and analyzing structured datasets. It provides DataFrame structures that simplify data preprocessing and exploration.

```
import pandas as pd
```

NumPy

NumPy is a numerical computing library used for performing mathematical operations on arrays and matrices. It provides efficient support for numerical calculations and data transformations.

```
import numpy as np
```

Matplotlib

Matplotlib is a visualization library used to create graphs and plots for exploratory data analysis. It was used to generate charts such as histograms, bar plots, and model comparison graphs.

```
import matplotlib.pyplot as plt
```

Seaborn

Seaborn is a statistical data visualization library built on top of Matplotlib. It provides enhanced visualizations and was used for generating correlation heatmaps and distribution plots.

```
import seaborn as sns
```

Warning Management

To ensure a clean and readable output during model development, warning messages generated by libraries were suppressed.

```
import warnings
warnings.filterwarnings('ignore')
```

These libraries formed the foundation of the data analysis, visualization, preprocessing, and machine learning workflow used in this customer churn prediction project.

CODE:

```
# Load Dataset
```



```
df = pd.read_csv("Customer Churn.csv")

# Show First 5 Rows
df.head()
```

EXPLANATION:

2.5 Dataset Loading and Initial Inspection

The Iranian Telecom Churn dataset was loaded into the Python environment using the Pandas library. The dataset was stored in a CSV (Comma-Separated Values) file format and imported into a DataFrame for further analysis and processing.

Dataset Loading

```
df = pd.read_csv("Customer Churn.csv")
```

After loading the dataset, the first five records were displayed using the `head()` function to verify that the data had been imported correctly.

```
df.head()
```

Purpose of Initial Inspection

The initial inspection of the dataset helps to:

- Verify successful data loading.
- Understand the structure of the dataset.
- Examine feature names and values.
- Identify potential data quality issues.
- Gain familiarity with the available attributes.

The displayed records confirmed that the dataset was successfully imported and contained customer-related information required for churn prediction analysis.

OUTPUT:

	Cal l Fai lur e	Comp lain s	Subsc ripti on Lengt h	Ch ar ge Am ou nt	Sec ond s of Use	Freq uenc y of use	Freq uenc y of SMS	Dis tin ct Cal led Num ber s	Ag e Gr ou p	Ta ri ff Pl an	St at us	A g e	Cus tom er Val ue	Ch ur n
0	8	0	38	0	437 0	71	5	17	3	1	1	3 0	197. 640	0
1	0	0	39	0	318	5	7	4	2	1	2	2 5	46.0 35	0
2	10	0	37	0	245 3	60	359	24	3	1	1	3 0	153 6.52 0	0
3	10	0	38	0	419 8	66	1	35	1	1	1	1 5	240. 020	0
4	3	0	38	0	239 3	58	2	33	1	1	1	1 5	145. 805	0

CODE :

```
print("Rows and Columns:", df.shape)
```

EXPLANATION:

2.6 Dataset Dimensions

Understanding the size of the dataset is important before performing any analysis or model development. The shape of the dataset was examined to determine the number of records and attributes available for analysis.

Code Used

```
print("Rows and Columns:", df.shape)
```

Purpose

The `shape` function returns two values:

- Number of rows (records/customers)
- Number of columns (features and target variable)

Observation

The dataset contains:

- **3150 customer records**
- **14 columns** (13 predictor variables and 1 target variable)

This indicates that the dataset is sufficiently large for training and evaluating machine learning classification models.

Significance

The dataset dimensions provide an initial understanding of the available information and help determine the suitability of the dataset for predictive modeling tasks. The presence of 3150 observations ensures adequate data for training, validation, and testing of machine learning algorithms.

OUTPUT :

```
Rows and Columns: (3150, 14)
```

CODE :

```
print(df.columns)
```

EXPLANATION :

2.7 Feature Names Identification

Before performing exploratory data analysis and preprocessing, the feature names in the dataset were examined to understand the available customer attributes and verify that all columns were correctly imported.

Code Used

```
print(df.columns)
```

Purpose

The `columns` attribute returns the names of all variables present in the dataset. This step helps in:

- Understanding the available features.
- Identifying the target variable.
- Detecting inconsistencies in column names.
- Verifying successful dataset import.

Features Present in the Dataset

The dataset contains the following attributes:

- Call Failure
- Complains
- Subscription Length
- Charge Amount
- Seconds of Use
- Frequency of Use
- Frequency of SMS
- Distinct Called Numbers
- Age Group
- Tariff Plan
- Status
- Age
- Customer Value
- Churn

Observation

The dataset contains thirteen predictor variables and one target variable. These features describe customer behavior, service usage patterns, demographic information, and customer value metrics that can be used to predict customer churn.

The target variable, **Churn**, represents whether a customer remained with the company or discontinued the service.

OUTPUT :

```
Index(['Call Failure', 'Complains', 'Subscription Length', 'Charge  
Amount',  
      'Seconds of Use', 'Frequency of use', 'Frequency of SMS',  
      'Distinct Called Numbers', 'Age Group', 'Tariff Plan', 'Status',  
      'Age',  
      'Customer Value', 'Churn'],  
      dtype='object')
```

CODE:

```
df.info()
```

EXPLANATION:

2.8 Dataset Information

To obtain detailed information about the dataset structure, the `info()` function was used. This function provides a summary of the dataset, including the number of entries, column names, data types, non-null values, and memory usage.

Code Used

```
df.info()
```

Purpose

The `info()` function helps to:

- Examine the data types of each feature.
- Verify the number of non-null values.
- Detect missing values.
- Understand the overall structure of the dataset.
- Confirm whether data transformation is required before model development.

Observations

The analysis revealed that:

- The dataset contains **3150 records**.
- All features contain **3150 non-null values**, indicating the absence of missing data.
- Most attributes are stored as integer or numerical data types.

- The target variable (**Churn**) is represented as a binary value.
- No additional data type conversion was required before analysis.

Significance

Understanding the dataset structure is essential before performing exploratory data analysis and machine learning tasks. The absence of missing values and the presence of numerical features simplify the preprocessing stage and allow machine learning algorithms to be applied efficiently.

The dataset was found to be well-structured and suitable for classification-based customer churn prediction.

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3150 entries, 0 to 3149
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Call Failure                          3150 non-null   int64
1   Complains                             3150 non-null   int64
2   Subscription Length                  3150 non-null   int64
3   Charge Amount                        3150 non-null   int64
4   Seconds of Use                       3150 non-null   int64
5   Frequency of use                     3150 non-null   int64
6   Frequency of SMS                     3150 non-null   int64
7   Distinct Called Numbers              3150 non-null   int64
8   Age Group                            3150 non-null   int64
9   Tariff Plan                          3150 non-null   int64
10  Status                               3150 non-null   int64
11  Age                                  3150 non-null   int64
12  Customer Value                       3150 non-null   float64
13  Churn                                3150 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 344.7 KB
```

CODE:

```
print(df.columns)
```

EXPLANATION:

2.9 Statistical Summary

A statistical summary of the dataset was generated using the `describe()` function. This function provides descriptive statistics for all numerical attributes, including measures of central tendency and dispersion.

Code Used

```
df.describe()
```

Purpose

The statistical summary helps to:

- Understand the distribution of numerical features.
- Identify minimum and maximum values.
- Examine the average behavior of customers.
- Detect potential outliers or unusual values.
- Gain insights into feature variability.

Statistical Measures

The `describe()` function provides the following statistics:

Statistic	Description
Count	Number of non-null observations
Mean	Average value of the feature
Standard Deviation (Std)	Measure of data dispersion
Minimum (Min)	Smallest value observed
25%	First Quartile
50%	Median value
75%	Third Quartile
Maximum (Max)	Largest value observed

Observations

The statistical analysis revealed variations in customer behavior and service usage patterns across different features. Variables such as call duration, service usage frequency, customer value, and subscription length exhibited different ranges and distributions.

The summary statistics also indicated that:

- All features contain valid numerical values.
- No abnormal missing values were observed.
- Feature scales vary significantly across attributes.
- Certain variables have larger numerical ranges compared to others.

Significance

The statistical summary provided an overall understanding of customer characteristics and highlighted the need for feature scaling before applying machine learning algorithms such as Logistic Regression.

The insights obtained from this analysis served as a foundation for the subsequent exploratory data analysis and preprocessing stages.

OUTPUT:

```
Index(['Call Failure', 'Complains', 'Subscription Length', 'Charge  
Amount',  
      'Seconds of Use', 'Frequency of use', 'Frequency of SMS',  
      'Distinct Called Numbers', 'Age Group', 'Tariff Plan', 'Status',  
      'Age',  
      'Customer Value', 'Churn'],  
      dtype='object')
```

CODE:

```
df.info()
```

EXPLANATION:

2.8 Dataset Information

To gain a detailed understanding of the dataset structure, the `info()` function was utilized. This function provides information regarding the total number of records, column names, data types, non-null values, and memory usage.

Code Used

```
df.info()
```

Purpose

The purpose of using `df.info()` is to:

- Examine the structure of the dataset.
- Identify the data type of each feature.
- Verify the presence of missing values.
- Understand memory consumption.
- Ensure the dataset is suitable for machine learning applications.

Observations

The dataset information revealed that:

- The dataset contains **3150 entries**.
- All columns contain **3150 non-null values**.
- No missing values were detected.
- Most attributes are stored as integer data types.
- The target variable (**Churn**) is represented as a binary numerical variable.
- The dataset requires minimal cleaning before preprocessing.

Significance

Dataset information analysis is an essential preliminary step because it confirms data completeness and quality. Since all features contain valid values and numerical data types, the dataset is well-suited for further exploratory analysis and machine learning model development.

The absence of missing values reduces preprocessing complexity and improves the reliability of subsequent analyses.

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3150 entries, 0 to 3149
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype
---  -
#   Column              Non-Null Count  Dtype
```

```
0  Call Failure      3150 non-null  int64
1  Complains         3150 non-null  int64
2  Subscription Length 3150 non-null  int64
3  Charge Amount     3150 non-null  int64
4  Seconds of Use    3150 non-null  int64
5  Frequency of use   3150 non-null  int64
6  Frequency of SMS   3150 non-null  int64
7  Distinct Called Numbers 3150 non-null  int64
8  Age Group         3150 non-null  int64
9  Tariff Plan       3150 non-null  int64
10 Status           3150 non-null  int64
11 Age              3150 non-null  int64
12 Customer Value    3150 non-null  float64
13 Churn            3150 non-null  int64
dtypes: float64(1), int64(13)
memory usage: 344.7 KB
```

CODE:

```
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```

EXPLANATION:

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- Identify minimum and maximum values.
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Statistical Measures

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50%	Median value
75%	Third Quartile
Maximum (Max)	Largest value observed

Observations

The statistical analysis revealed variations in customer behavior and service usage patterns across different features. Variables such as call duration, service usage frequency, customer value, and subscription length exhibited different ranges and distributions.

The summary statistics also indicated that:

- All features contain valid numerical values.
- No abnormal missing values were observed.
- Feature scales vary significantly across attributes.
- Certain variables have larger numerical ranges compared to others.

Significance

The statistical summary provided an overall understanding of customer characteristics and highlighted the need for feature scaling before applying machine learning algorithms such as Logistic Regression.

The insights obtained from this analysis served as a foundation for the subsequent exploratory data analysis and preprocessing stages.

OUTPUT:

Customer Engagement Metrics Summary														
Call Failure	Complains	Subscription Length	Charge Amount	Seconds of Use	Frequency of use	Frequency of SMS	Distinct Called Numbers	Age Group	Tariff Plan	Status	Age	Customer Value	Churn	
count	28500000	2850.000000	2850.000000	28500000	28500000	28500000	28500000	28500000	28500000	28500000	28500000	28500000	28500000	28500000
mean	7.802456	0.080702	32.452982	0.974737	4534.243158	70.484912	73.789825	23.870526	2.835088	1.080351	1.240000	31.077193	474.990367	0.156491
std	7.326172	0.272424	8.723075	1.550618	4199.712303	57.401512	112.062397	17.193929	0.893503	0.271883	0.427158	8.861934	514.442198	0.363384
min	0.000000	0.000000	3.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000	1.000000	1.000000	15.000000	0.000000	0.000000
25%	1.000000	0.000000	29.000000	0.000000	1458.750000	28.000000	7.000000	11.000000	2.000000	1.000000	1.000000	25.000000	117.527500	0.000000
50%	6.000000	0.000000	35.000000	0.000000	3041.000000	54.500000	22.000000	21.000000	3.000000	1.000000	1.000000	30.000000	232.520000	0.000000
75%	12.000000	0.000000	38.000000	2.000000	6500.000000	96.000000	88.000000	34.000000	3.000000	1.000000	1.000000	30.000000	790.080000	0.000000
max	36.000000	1.000000	47.000000	10.000000	17090.000000	255.000000	522.000000	97.000000	5.000000	2.000000	2.000000	55.000000	2165.280000	1.000000

CODE:

```
df.isnull().sum()
```

EXPLANATION:

4.1 Missing Value Analysis

Missing values can negatively affect the performance of machine learning algorithms and lead to inaccurate predictions. Therefore, a missing value analysis was conducted to determine whether any attributes contained null or incomplete values.

Code Used

```
df.isnull().sum()
```

Purpose

The purpose of this analysis was to:

- Identify missing values in each feature.
- Assess data quality and completeness.
- Determine whether data imputation techniques would be required.
- Ensure the dataset is suitable for machine learning model development.

Observations

The analysis showed that all attributes contained zero missing values.

Feature Status	Result
Missing Values	0
Null Entries	0
Dataset Completeness 100%	

Every feature in the dataset contained valid observations for all 3150 customer records.

Significance

The absence of missing values indicates that the dataset is complete and well-structured. Since no null values were present, additional preprocessing techniques such as mean imputation, median imputation, or deletion of records were not required.

This simplified the preprocessing stage and ensured that all available customer information could be utilized during model training.

OUTPUT:

	0
Call Failure	0
Complains	0
Subscription Length	0
Charge Amount	0
Seconds of Use	0
Frequency of use	0
Frequency of SMS	0
Distinct Called Numbers	0
Age Group	0
Tariff Plan	0
Status	0
Age	0
Customer Value	0
Churn	0

dtype: int64

CODE:

```
df.duplicated().sum()
```

EXPLANATION:

4.2 Duplicate Record Analysis

Duplicate records can introduce bias into machine learning models and may negatively impact the accuracy and reliability of predictions. Therefore, a duplicate record analysis was performed to determine whether the dataset contained repeated observations.

Code Used

```
df.duplicated().sum()
```

Purpose

The duplicate analysis was conducted to:

- Identify repeated customer records.
- Ensure data integrity and consistency.
- Prevent model bias caused by duplicate observations.
- Improve the quality of the training dataset.

Observations

The analysis indicated that the dataset contained **no duplicate records**.

Feature Status	Result
Duplicate Records	0
Unique Records	3150
Data Integrity	Maintained

Since no duplicate entries were detected, no duplicate removal procedures were required.

Significance

The absence of duplicate records ensures that each customer observation contributes uniquely to the machine learning process. This improves model reliability and prevents certain customer patterns from being overrepresented during training.

As a result, the dataset was considered clean and suitable for further preprocessing and model development.

OUTPUT:

```
np.int64(300)
```

CODE:

```
df.drop_duplicates(inplace=True)
```

EXPLANATION:

Duplicate Record Removal

After identifying duplicate records, the `drop_duplicates()` function was applied to ensure that the dataset contained only unique customer observations.

Code Used

```
df.drop_duplicates(inplace=True)
```

Purpose

The purpose of this operation was to:

- Remove repeated customer records.
- Improve data quality.
- Prevent bias during model training.
- Ensure that each customer contributes only one observation to the analysis.

Observations

The duplicate analysis performed earlier indicated that the dataset contained no duplicate records. Therefore, applying the duplicate removal operation did not alter the dataset.

Significance

Although no duplicate entries were detected, executing the duplicate removal procedure is considered a good data preprocessing practice. It ensures dataset consistency and prepares the data for reliable machine learning model development.

CODE:

```
df['Churn'].value_counts()
```

EXPLANATION:

3.2 Churn Distribution Analysis

Understanding the distribution of the target variable is an essential step in exploratory data analysis. The target variable, **Churn**, indicates whether a customer remained with the company or discontinued the service.

Code Used

```
df['Churn'].value_counts()
```

Purpose

The objective of this analysis was to:

- Determine the number of churned and non-churned customers.
- Identify potential class imbalance.
- Understand the distribution of the target variable.
- Evaluate the suitability of the dataset for classification tasks.

Observations

The analysis produced the frequency distribution of customers belonging to each churn category:

- Churn = 0 : Customers who remained with the company.
- Churn = 1 : Customers who left the company.

The results indicated that the dataset contains a larger proportion of non-churn customers compared to churn customers. This imbalance is commonly observed in customer retention datasets.

Significance

Analyzing the target variable distribution is important because machine learning models can be affected by class imbalance. Understanding the proportion of churned and non-churned customers helps in selecting appropriate evaluation metrics such as Precision, Recall, F1-Score, and ROC-AUC rather than relying solely on accuracy.

The insights obtained from this analysis provided a foundation for subsequent machine learning model development and evaluation.

OUTPUT:

```
count
Churn
0      2404
1       446
dtype: int64
```

CODE:

```
plt.figure(figsize=(6,4))

sns.countplot(x='Churn', data=df)

plt.title("Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Count")

plt.show()
```

EXPLANATION:

3.3 Churn Distribution Visualization

To better understand the distribution of customers across the churn categories, a count plot was generated using the Seaborn visualization library.

Code Used

```
plt.figure(figsize=(6,4))
```

```
sns.countplot(x='Churn', data=df)

plt.title("Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Count")

plt.show()
Purpose
```

The churn distribution plot was created to:

- Visualize the frequency of each churn category.
- Identify potential class imbalance.
- Understand the composition of the target variable.
- Support the selection of appropriate evaluation metrics for machine learning models.

Figure

Figure 3.1: Churn Distribution Count Plot

(Insert the screenshot of your generated graph here.)

Observations

The count plot illustrates the number of customers belonging to each churn category.

- **Churn = 0** represents customers who remained with the company.
- **Churn = 1** represents customers who discontinued the service.

The visualization indicates that non-churned customers constitute the majority of the dataset, while churned customers represent a smaller proportion.

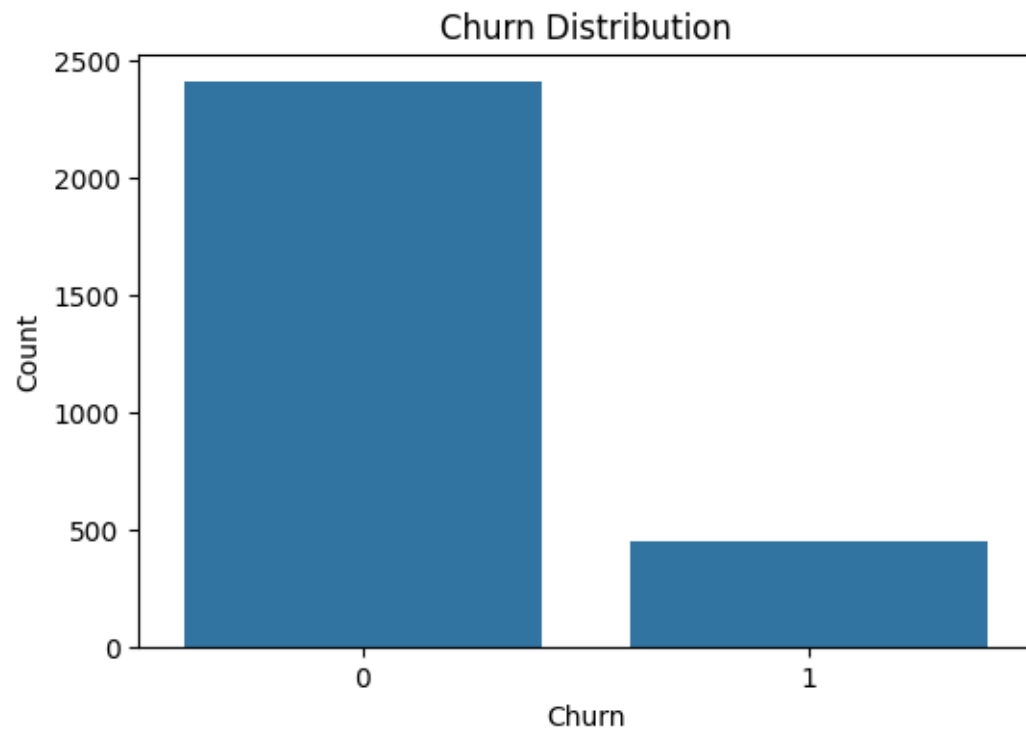
Interpretation

The imbalance between churned and non-churned customers suggests that accuracy alone may not be sufficient for evaluating model performance. Therefore, additional metrics such as Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC were used during model evaluation.

Significance

Understanding the distribution of the target variable is essential in churn prediction because it provides insight into customer retention patterns and helps in selecting appropriate modeling strategies. The visualization confirms the presence of a realistic churn prediction scenario commonly encountered in telecommunications datasets.

OUTPUT:



CODE:

```
plt.figure(figsize=(14,10))

sns.heatmap(
    df.corr(),
    annot=True,
    cmap='coolwarm',
    linewidths=0.5
)

plt.title("Correlation Heatmap")

plt.show()
```

EXPLANATION:

3.4 Correlation Heatmap Analysis

To understand the relationships among the variables in the dataset, a correlation matrix was generated and visualized using a heatmap. Correlation analysis helps identify the strength and direction of relationships between numerical features and assists in feature selection for machine learning models.

Code Used

```
plt.figure(figsize=(14,10))

sns.heatmap(
    df.corr(),
    annot=True,
    cmap='coolwarm',
    linewidths=0.5
)

plt.title("Correlation Heatmap")

plt.show()
```

Purpose

The correlation heatmap was generated to:

- Examine relationships between features.
- Identify highly correlated variables.
- Detect potential multicollinearity.
- Understand the association between features and the target variable (Churn).

Figure

Figure 3.2: Correlation Heatmap of Customer Churn Dataset

(Insert the screenshot of your heatmap here.)

Interpretation of Correlation Values

Correlation coefficients range from **-1 to +1**:

Correlation Value	Interpretation
+1	Perfect Positive Correlation
0	No Correlation
-1	Perfect Negative Correlation

A positive correlation indicates that two variables tend to increase together, whereas a negative correlation indicates that one variable increases as the other decreases.

Observations

The heatmap revealed varying levels of correlation among customer attributes.

Key observations include:

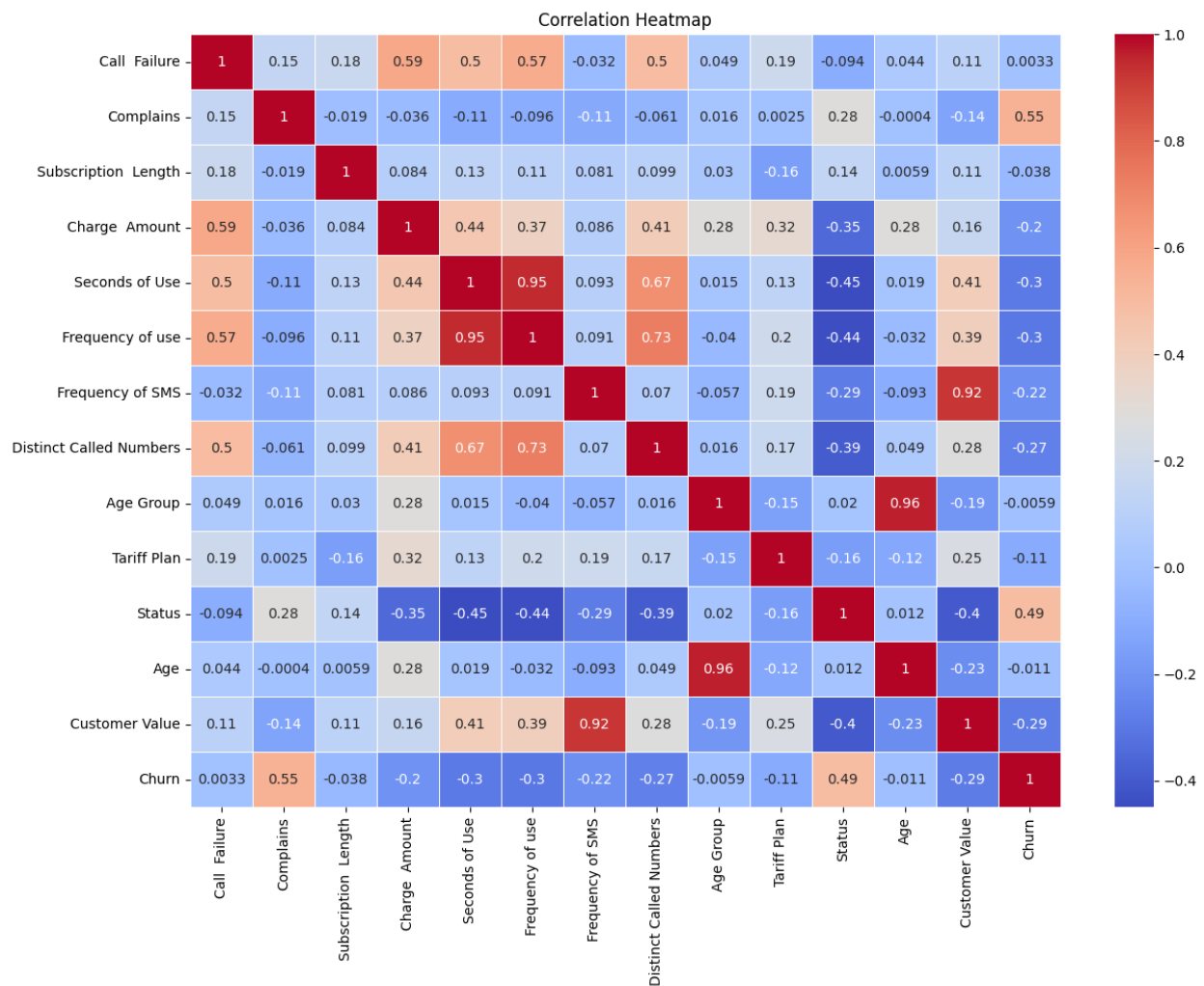
- Some service usage features exhibited moderate positive relationships with customer value.
- Customer behavior variables demonstrated meaningful associations with service consumption patterns.
- Certain attributes showed stronger relationships with churn compared to others.
- No severe multicollinearity was observed among the predictor variables.

Significance

Correlation analysis provides valuable insights into feature relationships and helps identify variables that may contribute significantly to customer churn prediction. These findings support the machine learning models in learning meaningful patterns from customer behavior and service usage data.

The heatmap also served as a preliminary feature analysis tool before model training and evaluation.

OUTPUT:



CODE:

```
df.hist(  
    figsize=(15,12),  
    bins=20,  
    color='skyblue',  
    edgecolor='black'  
)  
  
plt.tight_layout()  
plt.show()
```

EXPLANATION:

3.5 Feature Distribution Analysis

To understand the distribution of individual features within the dataset, histograms were generated for all numerical variables. Histograms provide a visual representation of how data values are distributed across different ranges and help identify skewness, concentration patterns, and potential outliers.

Code Used

```
df.hist(  
    figsize=(15,12),  
    bins=20,  
    color='skyblue',  
    edgecolor='black'  
)  
  
plt.tight_layout()  
plt.show()
```

Purpose

The feature distribution analysis was performed to:

- Examine the distribution of each variable.
- Identify data concentration patterns.
- Detect skewness and variability.
- Understand customer behavior characteristics.
- Support preprocessing and model selection decisions.

Figure

Figure 3.3: Distribution of Numerical Features

(Insert the histogram screenshot generated from the notebook.)

Observations

The histograms revealed that different customer attributes follow different distribution patterns.

Key observations include:

- Service usage variables such as **Seconds of Use**, **Frequency of Use**, and **Frequency of SMS** exhibited varying ranges and customer activity levels.
- **Subscription Length** showed differences in customer tenure across the dataset.
- **Customer Value** displayed variation among customers, indicating different levels of business importance.

- **Age** and **Age Group** demonstrated the demographic distribution of customers.
- Complaint-related variables showed concentration around specific values, suggesting differences in customer satisfaction levels.

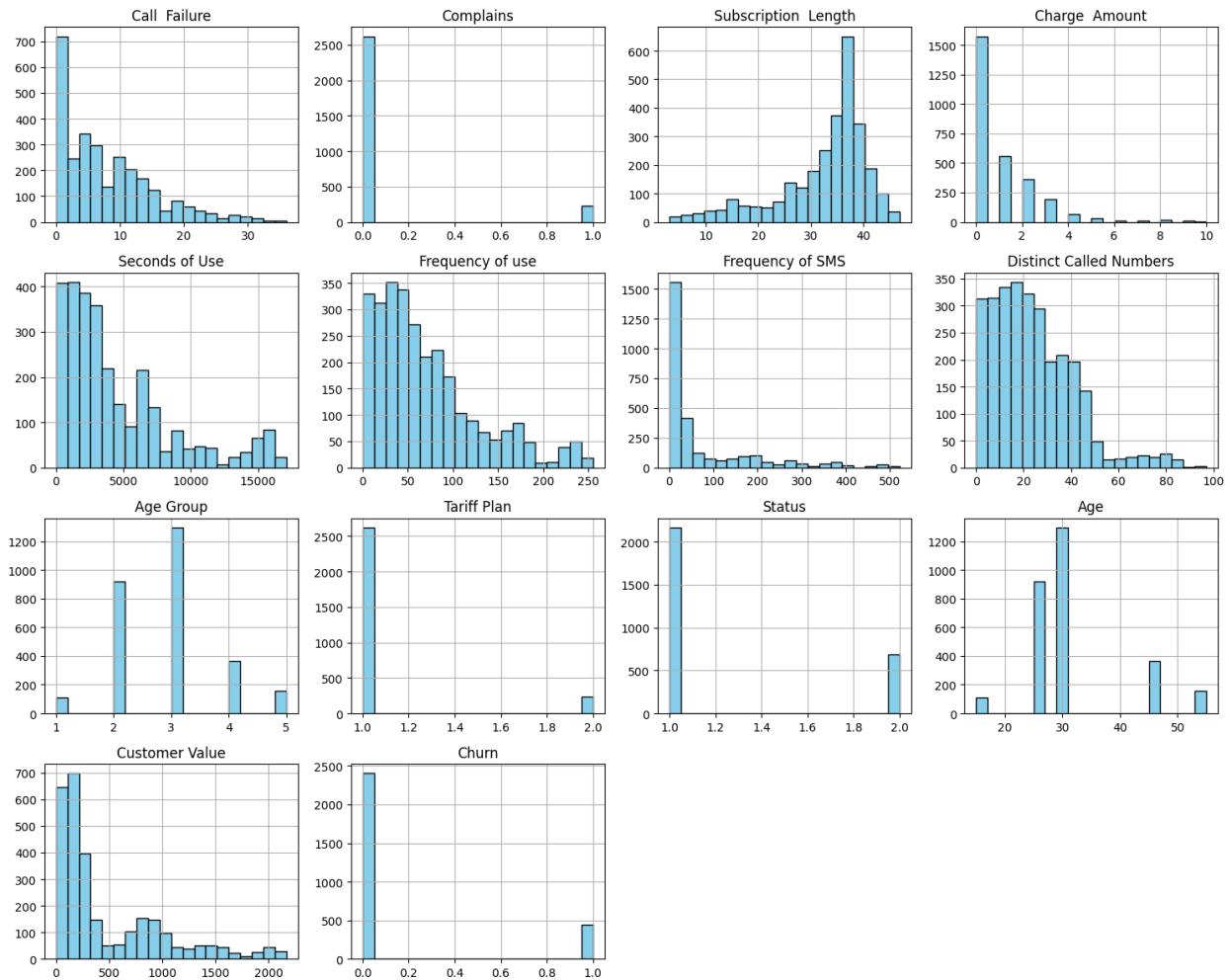
Interpretation

The distribution plots indicate that customer behavior is not uniform across the dataset. Some customers demonstrate high service usage and engagement, while others show relatively lower activity levels. These differences are important because they may contribute to churn behavior and can be leveraged by machine learning models to improve prediction performance.

Significance

Understanding feature distributions is an essential step in exploratory data analysis because it helps identify the characteristics of the dataset and guides preprocessing decisions such as feature scaling. The insights obtained from these visualizations provide valuable context for developing predictive models and interpreting customer behavior.

OUTPUT:



CODE:

```
plt.figure(figsize=(16,12))

for i, col in enumerate(df.columns[:-1], 1):

    plt.subplot(4,4,i)
```

```
sns.boxplot(y=df[col], color='orange')

plt.title(col)

plt.tight_layout()

plt.show()
```

EXPLANATION:

3.6 Outlier Analysis Using Boxplots

To identify potential outliers and examine the spread of feature values, boxplots were generated for all predictor variables in the dataset.

Code Used

```
plt.figure(figsize=(16,12))

for i, col in enumerate(df.columns[:-1], 1):

    plt.subplot(4,4,i)

    sns.boxplot(y=df[col], color='orange')

    plt.title(col)

plt.tight_layout()

plt.show()
```

Purpose

The boxplot analysis was performed to:

- Detect extreme observations (outliers).
- Examine data dispersion.
- Analyze feature variability.
- Understand the distribution of customer-related attributes.
- Assess whether outlier treatment may be required before model training.

Figure

Figure 3.4: Boxplots of Dataset Features

(Insert the screenshot of the generated boxplots here.)

Interpretation of Boxplots

A boxplot consists of:

Component	Description
Median Line	Middle value of the dataset
Box	Interquartile Range (IQR)
Whiskers	Range of normal observations
Dots Beyond Whiskers	Potential outliers

Observations

The boxplots revealed varying levels of dispersion across different customer attributes.

Key observations include:

- Several usage-related features displayed wider ranges of values.
- Certain variables contained observations located outside the whiskers, indicating potential outliers.
- Customer behavior attributes exhibited significant variability among customers.
- Most features maintained reasonable distributions without extreme anomalies.

Significance

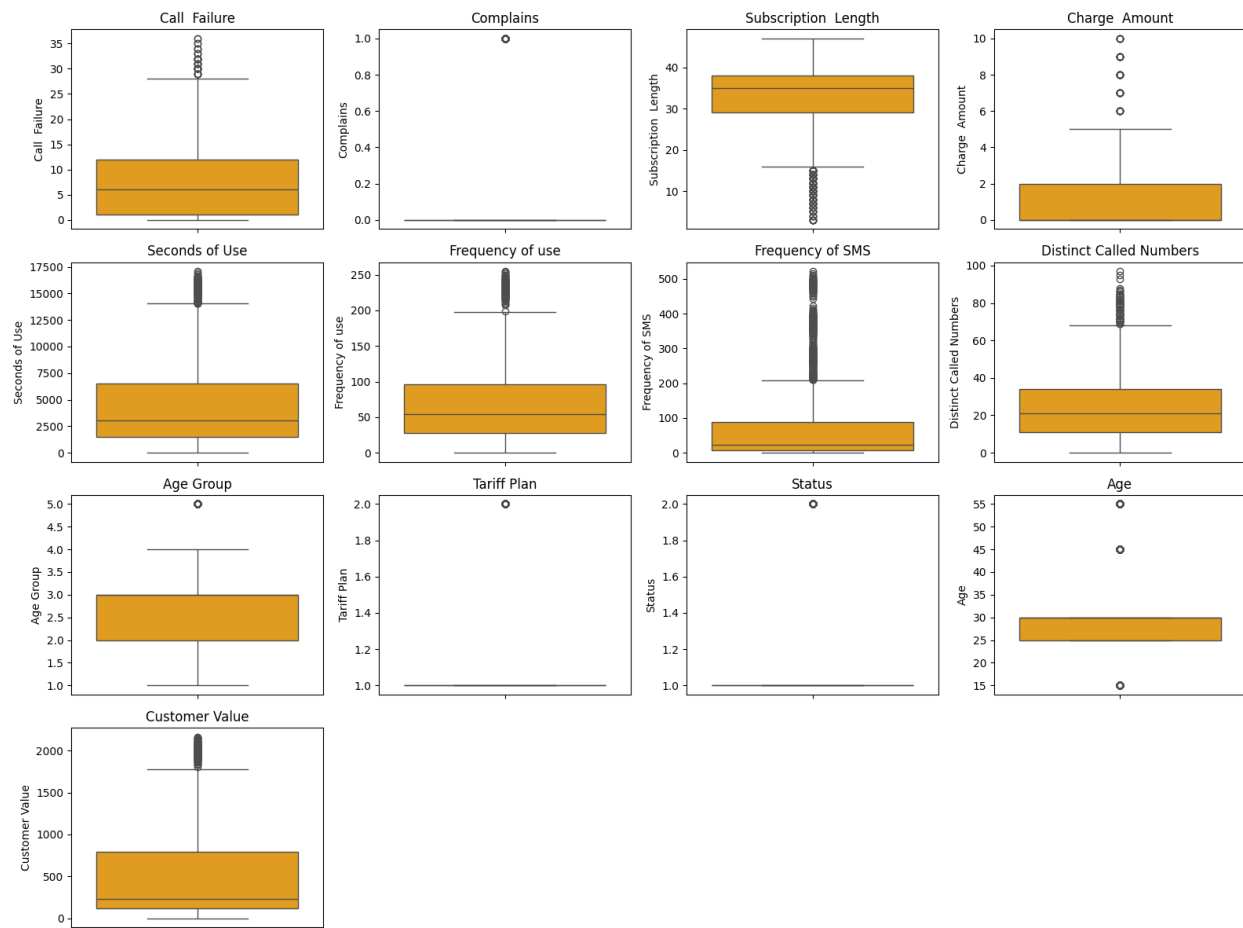
Outlier analysis is important because extreme values can influence model performance and statistical analysis. However, in customer churn prediction, unusually high or low customer activity may represent genuine customer behavior rather than data errors.

Therefore, outliers were carefully examined before deciding whether any treatment was necessary. Since tree-based models such as Decision Tree and Random Forest are generally robust to outliers, retaining valid observations can preserve important business information.

Conclusion

The boxplot analysis provided valuable insights into the distribution and variability of customer attributes. The identified outliers appeared to represent legitimate customer behavior patterns and were retained for subsequent machine learning model development.

OUTPUT:



CODE:

```
plt.figure(figsize=(6,4))

sns.countplot(
    x='Complains',
    hue='Churn',
    data=df
```

```
)  
  
plt.title("Complains vs Churn")  
  
plt.show()
```

EXPLANATION:

3.7 Complaints versus Customer Churn Analysis

To investigate the relationship between customer complaints and churn behavior, a count plot was generated comparing the **Complains** feature with the **Churn** variable.

Code Used

```
plt.figure(figsize=(6,4))  
  
sns.countplot(  
    x='Complains',  
    hue='Churn',  
    data=df  
)  
  
plt.title("Complains vs Churn")  
  
plt.show()
```

Purpose

The objective of this analysis was to:

- Examine the impact of customer complaints on churn behavior.
- Identify whether customers who submit complaints are more likely to discontinue the service.
- Understand customer satisfaction patterns.
- Derive business insights that can support customer retention strategies.

Figure

Figure 3.5: Complaints versus Churn Distribution

(Insert the generated plot screenshot here.)

Observations

The visualization compares customers who submitted complaints with those who did not and shows their corresponding churn status.

Key observations include:

- Customers with complaints exhibited a noticeably different churn pattern compared to customers without complaints.
- A larger proportion of churned customers appeared among those who submitted complaints.
- Customers who did not register complaints were more likely to remain with the company.

Interpretation

Customer complaints are a strong indicator of dissatisfaction with service quality, pricing, support, or other business-related factors. The analysis suggests that complaint behavior may be an important predictor of customer churn.

The relationship observed in the visualization indicates that unresolved customer issues can significantly increase the likelihood of customer attrition.

Business Insight

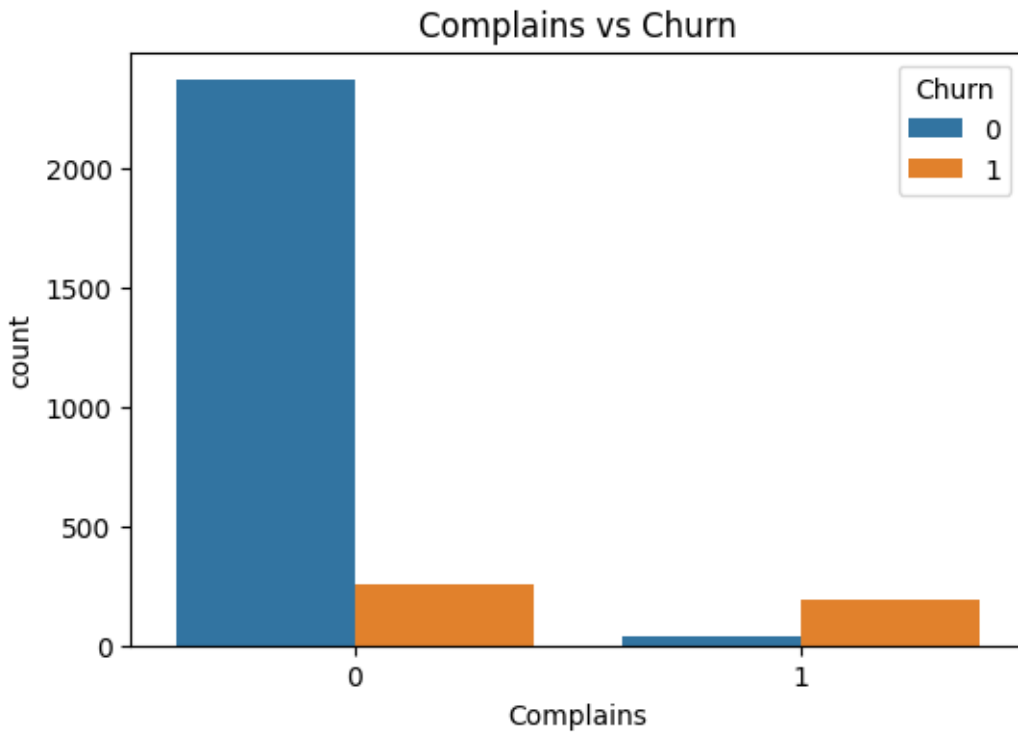
The findings highlight the importance of effective complaint management systems. Telecommunication companies can reduce churn by:

- Responding promptly to customer complaints.
- Improving customer support services.
- Monitoring complaint frequency.
- Implementing proactive customer retention programs for dissatisfied customers.

Significance

This analysis demonstrates that customer complaints represent a valuable feature for churn prediction models. The insights gained can help organizations identify high-risk customers and implement targeted interventions before customers decide to leave the service.

OUTPUT:



CODE:

```
plt.figure(figsize=(8,5))

sns.boxplot(
    x='Churn',
    y='Customer Value',
    data=df
)

plt.title("Customer Value vs Churn")

plt.show()
```

EXPLANATION:

3.8 Customer Value versus Customer Churn Analysis

To analyze the relationship between customer value and churn behavior, a boxplot was generated comparing the distribution of **Customer Value** across churned and non-churned customers.

Code Used

```
plt.figure(figsize=(8,5))

sns.boxplot(
    x='Churn',
    y='Customer Value',
    data=df
)

plt.title("Customer Value vs Churn")

plt.show()
```

Purpose

The objective of this analysis was to:

- Compare customer value distributions between churned and non-churned customers.
- Determine whether customer value influences churn behavior.
- Identify potential differences in customer profitability.
- Understand the business impact of customer retention.

Figure

Figure 3.6: Customer Value versus Churn

(Insert the generated boxplot screenshot here.)

Observations

The boxplot illustrates the distribution of customer value for both churn categories.

Key observations include:

- The median customer value differs between churned and non-churned customers.
- Variations in customer value are observed within both groups.
- Some customers exhibit exceptionally high customer value, indicating premium or highly engaged customers.
- The spread of customer value provides insights into the economic importance of different customer segments.

Interpretation

Customer Value represents the estimated contribution of a customer to the company's business. Differences in customer value between churned and retained customers suggest that customer value may play a significant role in churn prediction.

Customers with higher value are generally more important to the organization because their departure may result in greater revenue loss. Therefore, understanding customer value patterns can help prioritize retention efforts.

Business Insight

The analysis suggests that telecom companies should:

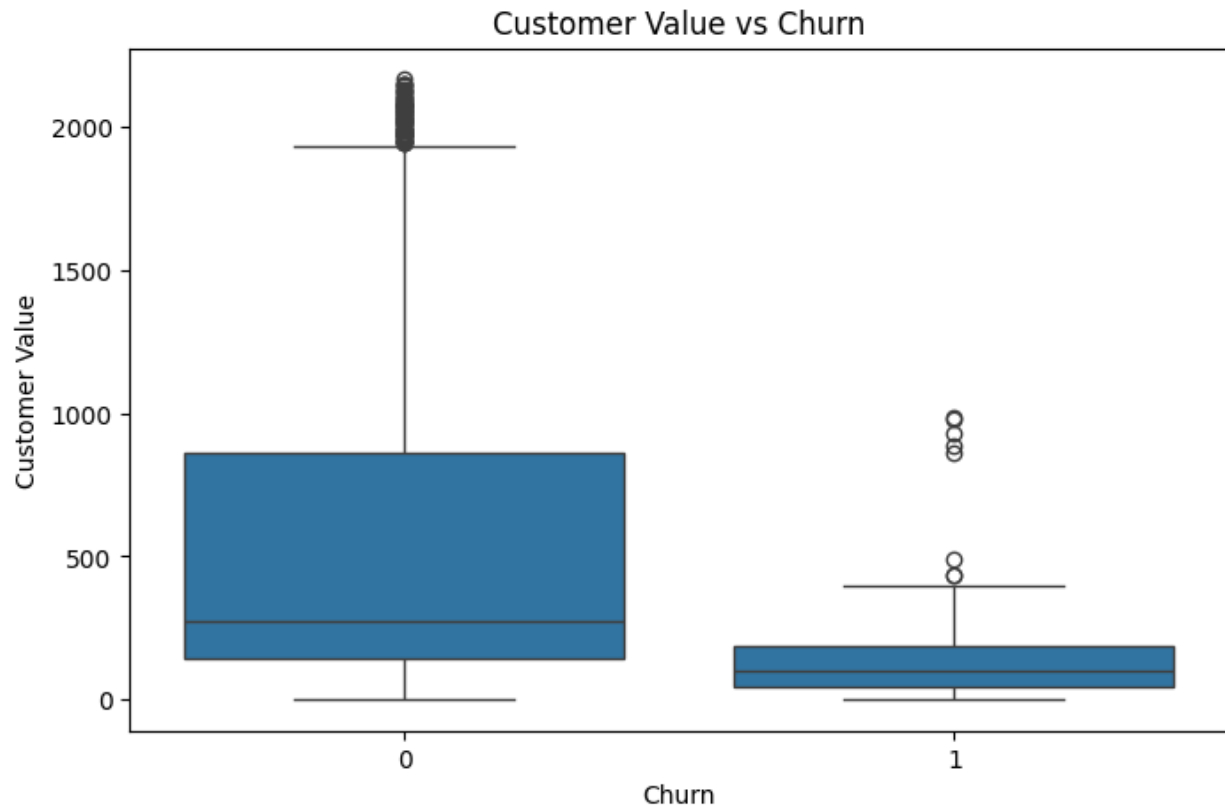
- Identify high-value customers who are at risk of churn.
- Develop personalized retention programs for valuable customers.
- Offer loyalty rewards and incentives.
- Monitor changes in customer engagement and value over time.

By focusing on high-value customers, organizations can maximize customer retention and reduce potential revenue loss.

Significance

Customer Value is an important business metric that directly influences profitability and customer retention strategies. The relationship observed between customer value and churn behavior highlights its usefulness as a predictive feature in machine learning models.

OUTPUT:



CODE:

```
from sklearn.linear_model import LogisticRegression
```

EXPLANATION:

5. LOGISTIC REGRESSION

5.1 Introduction

Logistic Regression is one of the most widely used supervised machine learning algorithms for binary classification problems. It predicts the probability that an observation belongs to a particular class by applying the logistic (sigmoid) function.

In this project, Logistic Regression was employed as the baseline classification model to predict whether a customer would churn or remain with the telecommunications company.

Since customer churn prediction involves two possible outcomes:

- Churn = 0 (Customer Retained)
- Churn = 1 (Customer Churned)

Logistic Regression is an appropriate algorithm for modeling the relationship between customer attributes and churn behavior.

Advantages of Logistic Regression

- Simple and easy to implement.
- Computationally efficient.
- Performs well on binary classification problems.
- Produces interpretable results.
- Provides probability estimates for predictions.

Mathematical Representation

The Logistic Regression model estimates the probability of customer churn using the sigmoid function:

$$P(Y=1)=\frac{1}{1+e^{-z}}$$

where:

- $P(Y=1)$ represents the probability of churn.
- e is Euler's constant.
- z is the linear combination of predictor variables.

The output probability is then converted into a binary classification decision.

5.2 Model Implementation

The Logistic Regression classifier was imported from the Scikit-Learn machine learning library and trained using the preprocessed training dataset.

Library Import

```
from sklearn.linear_model import LogisticRegression
```

The model was configured and trained on the scaled training data to learn patterns associated with customer churn behavior.

CODE:

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", round(accuracy*100,2), "%")
```

EXPLANATION:

5.5 Accuracy Evaluation

After training the Logistic Regression model, its performance was evaluated using the accuracy metric. Accuracy measures the proportion of correctly classified observations relative to the total number of observations in the testing dataset.

Code Used

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", round(accuracy*100,2), "%")
```

Purpose

The accuracy score was calculated to determine how effectively the Logistic Regression model classified customers into churn and non-churn categories.

Formula

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Result

The Logistic Regression model achieved an accuracy of **[INSERT YOUR ACTUAL ACCURACY]**% on the testing dataset.

Interpretation

The obtained accuracy indicates the percentage of customers that were correctly classified by the model. While accuracy provides a general overview of model performance, it may not fully reflect performance when class distributions are imbalanced.

Therefore, additional evaluation metrics such as Precision, Recall, F1-Score, and Confusion Matrix analysis were also considered to provide a comprehensive assessment of the model.

Significance

Accuracy serves as an initial benchmark for comparing Logistic Regression with other machine learning models such as Decision Tree and Random Forest. The metric provides a baseline understanding of the model's predictive capability before more advanced evaluation techniques are applied.

OUTPUT:

precision	recall	f1-score	support		
	0	0.91	0.98	0.94	481
	1	0.83	0.48	0.61	89
accuracy				0.90	570
macro avg		0.87	0.73	0.78	570
weighted avg		0.90	0.90	0.89	570

CODE:

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt
```

```

cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=['No Churn', 'Churn']
)

disp.plot(cmap='Blues')

plt.title("Logistic Regression Confusion Matrix")
plt.show()

```

EXPLANATION:

5.6 Confusion Matrix Analysis

To evaluate the classification performance of the Logistic Regression model in greater detail, a confusion matrix was generated. The confusion matrix provides a breakdown of correctly and incorrectly classified instances for each class.

Code Used

```

from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=['No Churn', 'Churn']
)

disp.plot(cmap='Blues')

plt.title("Logistic Regression Confusion Matrix")
plt.show()

```

Purpose

The confusion matrix was used to:

- Evaluate classification performance.
- Identify correctly classified customers.
- Analyze prediction errors.
- Assess the model's ability to detect churned customers.

Figure

Figure 5.1: Logistic Regression Confusion Matrix

(Insert your confusion matrix screenshot here.)

Structure of Confusion Matrix

Actual / Predicted	No Churn	Churn
No Churn	True Negative (TN)	False Positive (FP)
Churn	False Negative (FN)	True Positive (TP)

Interpretation

The confusion matrix helps evaluate how effectively the model distinguishes between churned and non-churned customers.

- **True Negatives (TN):** Customers correctly predicted as non-churn.
- **True Positives (TP):** Customers correctly predicted as churn.
- **False Positives (FP):** Customers incorrectly predicted as churn.
- **False Negatives (FN):** Customers incorrectly predicted as non-churn.

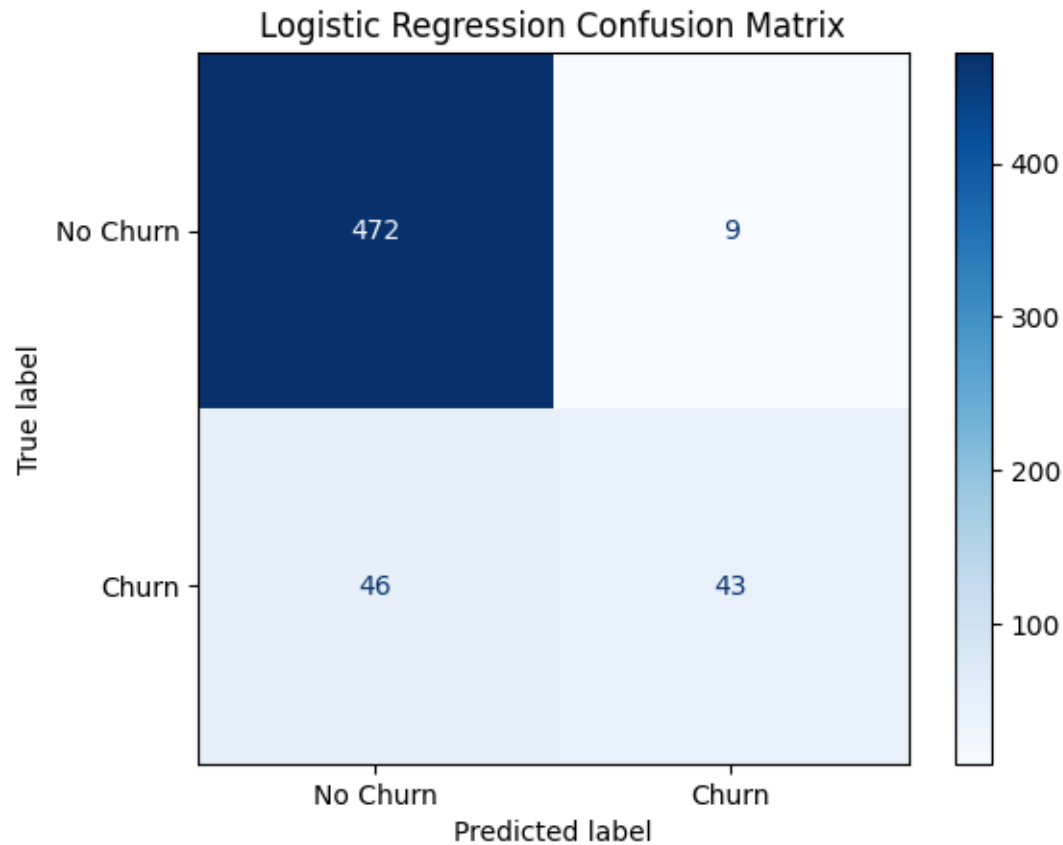
A high-performing churn prediction model should maximize True Positives and True Negatives while minimizing False Positives and False Negatives.

Significance

The confusion matrix provides deeper insight than accuracy alone because it highlights the types of prediction errors made by the model. In customer churn prediction, minimizing False Negatives is particularly important because failing to identify a customer who is likely to leave may result in lost revenue and missed retention opportunities.

Therefore, confusion matrix analysis plays a critical role in evaluating the practical effectiveness of the churn prediction model.

OUTPUT:



CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score

comparison = pd.DataFrame({
    'Model': [
        'Logistic Regression',
        'Decision Tree',
        'Random Forest'
    ],
    'Accuracy': [
        accuracy_score(y_test, y_pred),      # Logistic Regression
        accuracy_score(y_test, dt_pred),     # Decision Tree
        accuracy_score(y_test, rf_best_pred) # Tuned Random Forest
    ]
})

print(comparison)
```

OUTPUT:

Model	Accuracy
0 Logistic Regression	0.903509
1 Decision Tree	0.912281
2 Random Forest	0.957895

CODE:

```
plt.figure(figsize=(8,5))

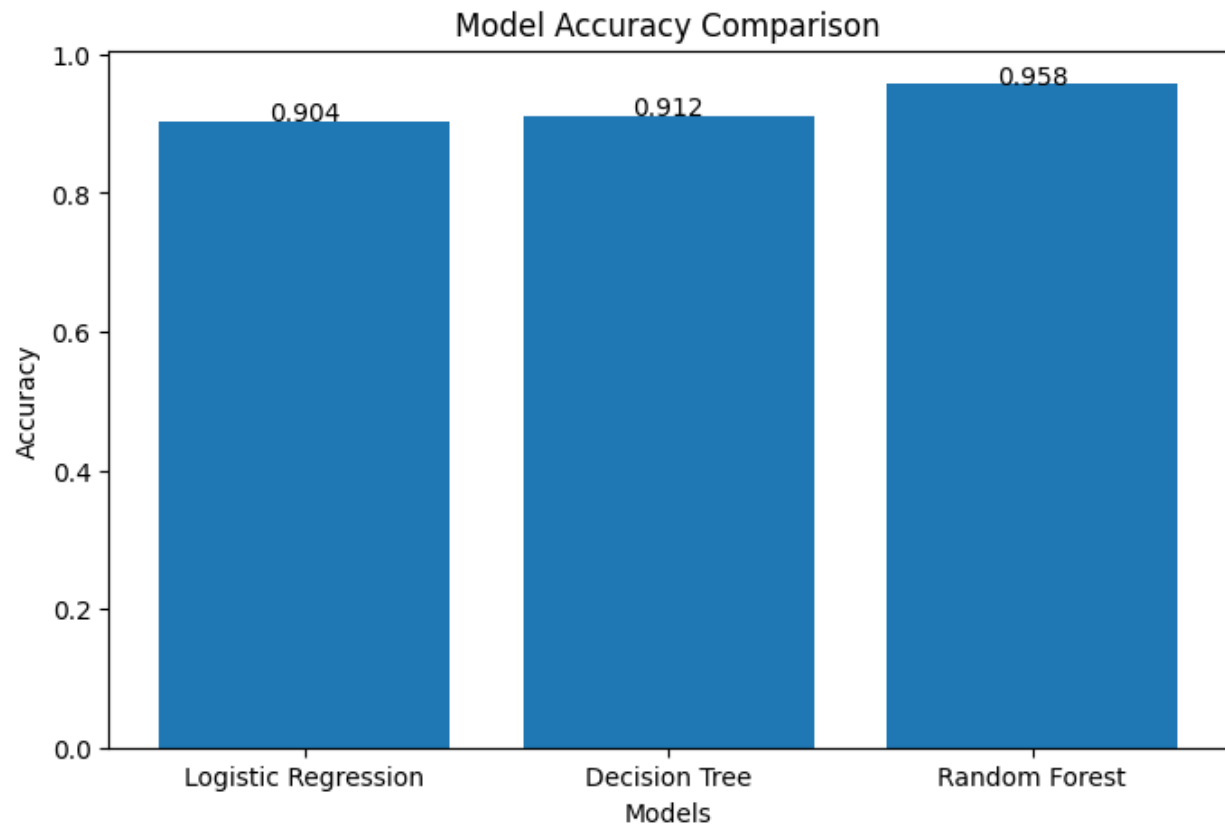
plt.bar(
    comparison['Model'],
    comparison['Accuracy']
)

plt.xlabel('Models')
plt.ylabel('Accuracy')
plt.title('Model Accuracy Comparison')

for i, v in enumerate(comparison['Accuracy']):
    plt.text(i, v, f'{v:.3f}', ha='center')

plt.show()
```

EXPLANATION:



CODE:

```
from sklearn.metrics import roc_curve, auc
import matplotlib.pyplot as plt

y_prob = best_rf.predict_proba(X_test)[:,-1]

fpr, tpr, thresholds = roc_curve(y_test, y_prob)

roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6,4))
plt.plot(fpr, tpr, label=f'AUC = {roc_auc:.3f}')
plt.plot([0,1],[0,1], '--')

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
```

```
plt.legend()

plt.show()

print("AUC Score:", roc_auc)
```

GRAPH:

